MediCore

Hospital Management System

A Full Stack Hospital Management System using Oracle Database

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Why MediCore? — From Manual Chaos to Digital Healthcare

1 What is MediCore?

→ Centralized HMS Platform

A web-based system managing Patients, Doctors, Lab Technicians and Administrators from one Oracle-backed database.

→ Objectives

Digitize appointments, prescriptions, lab reports, and financial ledgers. Eliminate paper-based workflows entirely.

→ Expected Impact

Faster patient service, zero data loss, transparent financials, and role-based secure access for all users.

2 Problems in Traditional Hospitals

Manual Paper Records

Patient files, prescriptions, and reports are stored on paper, making records hard to organize and retrieve.

Data Silos

Departments work in isolated systems, so information is not centralized or shared in real time.

High Error Rates

Handwritten entries and repetitive manual work increase the risk of mistakes and missing information.

Slow Patient Service

Searching files, updating records, and coordinating care takes too long for staff and patients.

Who Uses MediCore & How It's Built

1 Four User Roles & Their Features

Patient

- Book appointments
- View prescriptions
- Access lab reports
- Online payment

Doctor

- View appointments
- Write prescriptions
- Order lab tests
- Earnings & withdrawal

Lab Technician

- View pending tests
- Enter test results
- Upload reports

Administrator

- Manage doctors & depts
- Financial oversight
- System configuration

2 Technology Stack

Frontend

Next.js · React · Tailwind CSS

Backend

Node.js · Express.js · REST API

Database

Oracle Database 11g · PL/SQL · Sequences · Triggers

Auth & Payment

JWT · bcrypt · SSLCommerz

ARCHITECTURE & DATABASE DESIGN

System Architecture Overview

1 Multi-Tier System Flow

Client Tier

Next.js & React
Tailwind CSS UI

⇄

Application Tier

Express.js API
JWT & Business Logic

⇄

Database Tier

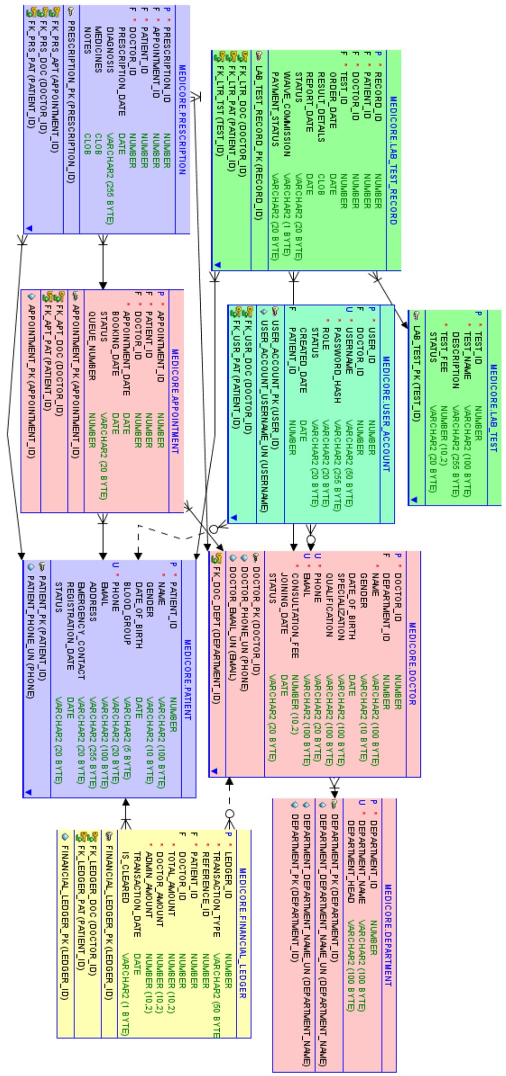
Oracle Database 11g
PL/SQL Sequences

REST API connects the frontend and backend. JWT tokens secure every request. Oracle handles all persistence.

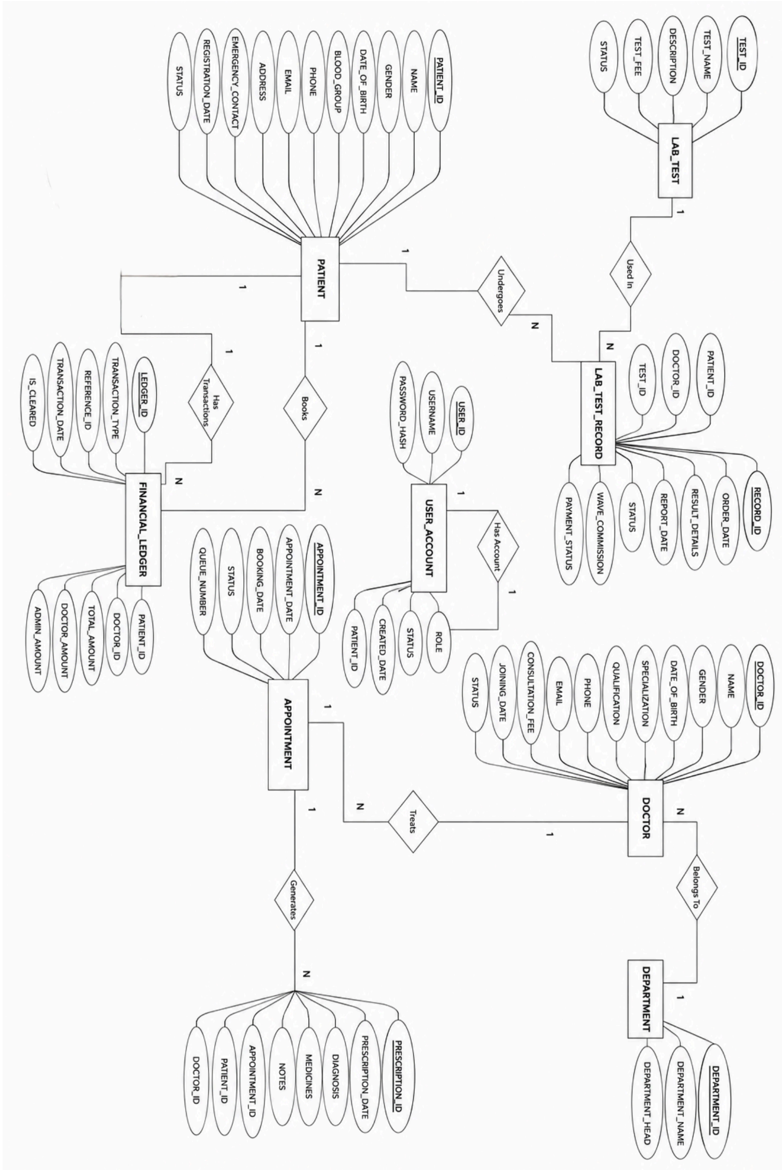
2 Database Entities

- **9 Tables** — Patient, Doctor, Department, Appointment, Prescription, Lab_Test, Lab_Test_Record, User_Account, Financial_Ledger
- Normalized to **3NF** with proper PK/FK relationships
- Oracle Sequences for auto-increment IDs
- Triggers for audit & commission calculation

Relational Database Schema



Entity Relationship Diagram



Key System Workflows

1 Authentication

Login (Enter credentials) → Role Access (Patient/Doctor/Admin) → JWT Token (Signed with secret) → Dashboard (Load role-specific view)

2 Appointment Booking

Select Doctor (Choose doctor/date) → Generate Queue (Oracle Sequence) → Process Payment (SSLCommerz) → Confirm Appt (Stored in database)

3 Prescription

Consultation (Patient attends appt) → Digital Rx (Created and saved) → Store in Oracle (CLOB data type) → Patient Dash (Views prescription)

4 Financial (80/20)

Payment (Collected online) → Revenue Split (80% Doctor/20% Hosp) → Ledger Update (Updated in Oracle) → Withdrawal (Doctor processed)

The Oracle-based queue algorithm uses sequences and PL/SQL triggers to automatically assign, track, and manage appointment queue numbers — ensuring fairness and no double-booking.

SECURITY FEATURES

Security Architecture

JWT Authentication

Stateless token-based auth. Every API request verified with signed JWT. Tokens expire to prevent session hijacking.

Route Protection

Backend middleware validates JWT and role on every protected route. Unauthorized requests are rejected with 401/403.

bcrypt Password Hashing

Passwords stored as salted bcrypt hashes. Plain-text passwords never stored in the Oracle database.

Role-Based Authorization

Four distinct roles — Patient, Doctor, Lab, Admin — each with strictly scoped access to API endpoints and UI views.

SQL Injection Prevention

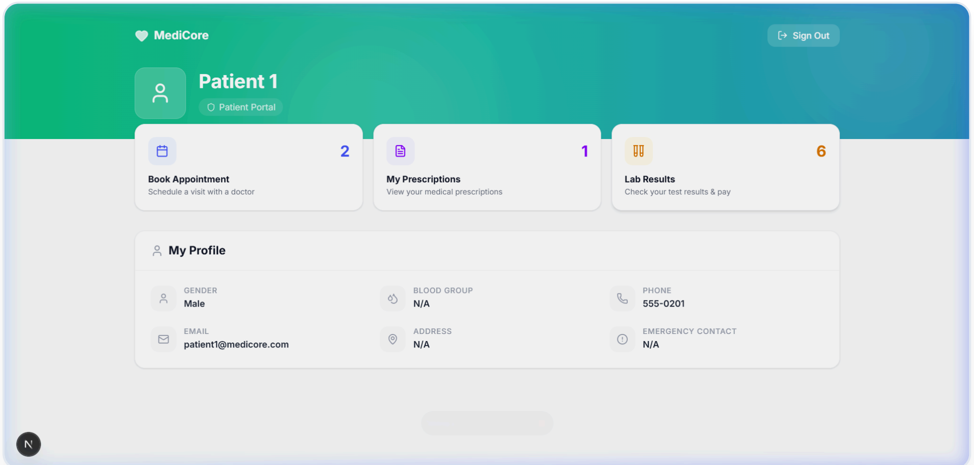
All Oracle queries use bind variables (parameterized queries), completely eliminating SQL injection attack vectors.

XSS Protection

Input sanitization on both frontend (React escaping) and backend layers prevents cross-site scripting attacks.

UI DEMONSTRATION

Patient Dashboard



1 Role & Access

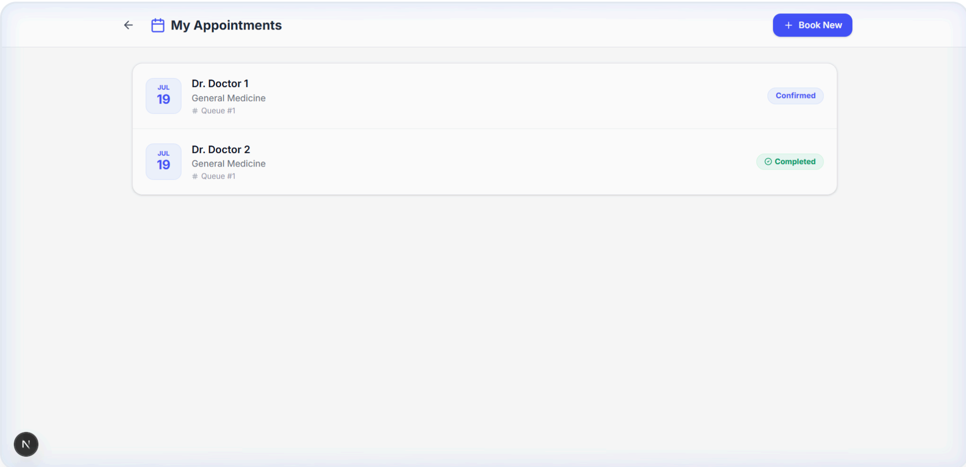
Primary Patient View

The central hub for authenticated patients to manage their healthcare records and navigate the system securely.

2 Key Features Displayed

- **Quick Stats Bar:** Shows total appointments, prescriptions, and lab results at a glance.
- **Patient Profile:** Medical profile data stored securely in the PATIENT table.
- **Secure Navigation:** Route-protected links to bookings and records.

Patient Appointments



1 Role & Access

Queue Tracking

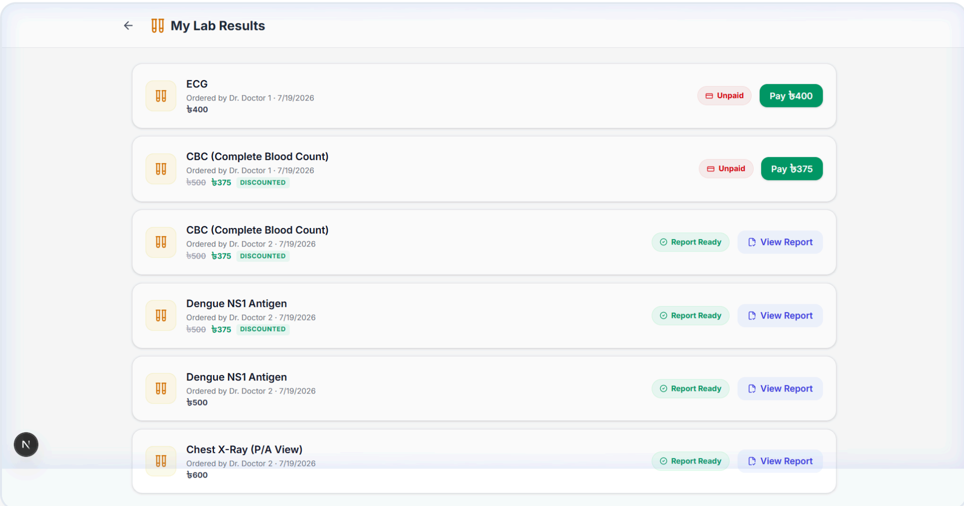
Allows patients to book new doctors and track their ongoing live queues.

2 Key Features Displayed

- **Appointment Status:** Patients see confirmed/completed status badges per doctor.
- **Live Queue Number:** Queried dynamically via PL/SQL sequences to avoid race conditions.
- **Online Booking:** Integration with SSLCommerz for booking fee.

UI DEMONSTRATION

Patient Lab Results & Prescriptions



1 Role & Access

Digital Health Records

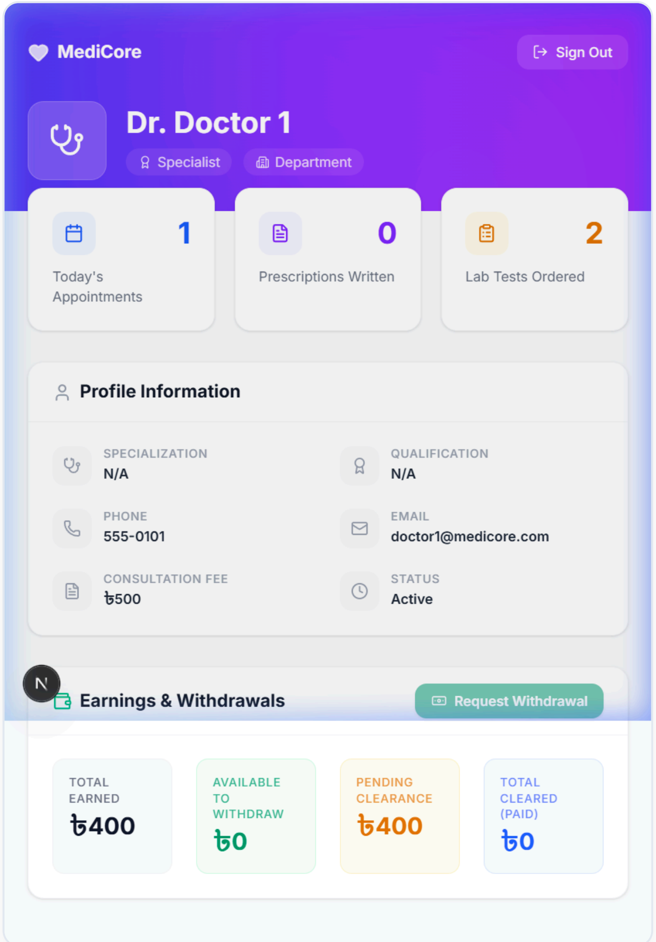
Gives patients permanent online access to their historical diagnostic and consultation data.

2 Key Features Displayed

- **Digital Prescriptions:** View doctors' notes instantly.
- **Secure Lab Reports:** Download finalized test results in PDF format directly from Oracle CLOBs.
- **Pending Actions:** Alerts for unpaid lab tests ordered by doctors.

UI DEMONSTRATION

Doctor Panel



1 Role & Access

Physician Dashboard

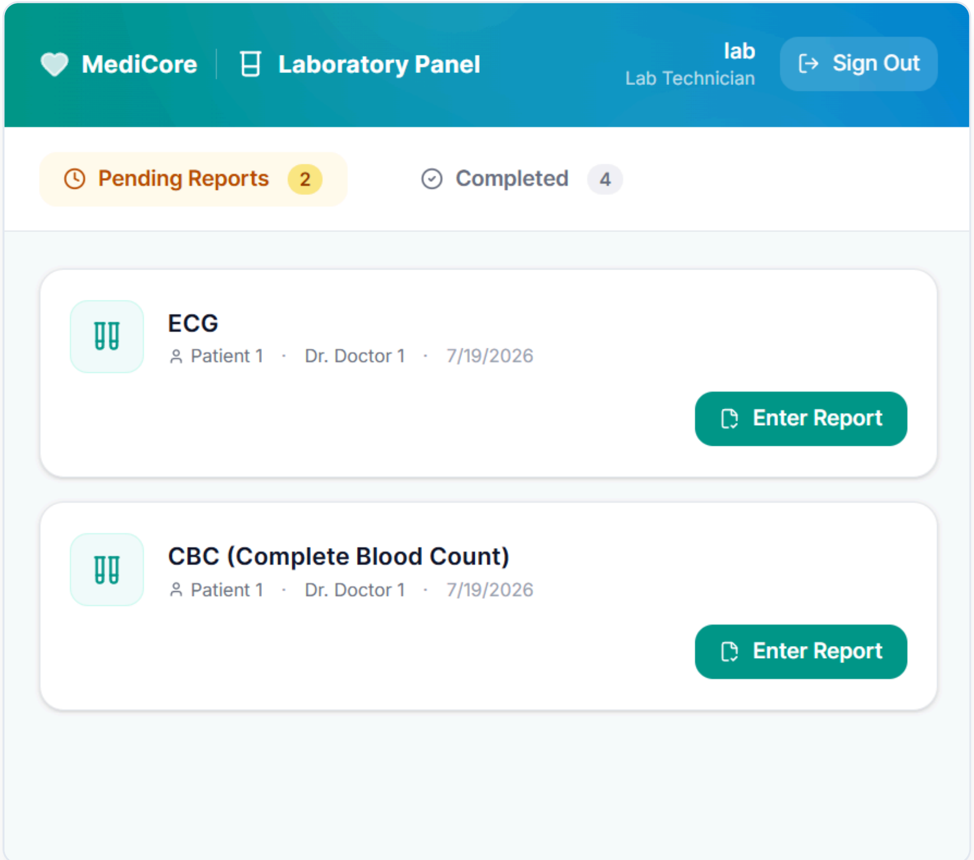
Exclusive workspace for doctors to manage patient flow and track their revenue.

2 Key Features Displayed

- **Appointments Count:** See total patients lined up for the day.
- **Prescriptions Written:** Track total consultations completed.
- **Financial Balance:** View pending, cleared, and total earnings based on the 80% commission split.
- **Withdrawal Requests:** Trigger payouts directly to the admin queue.

UI DEMONSTRATION

Laboratory Operations Panel



1 Role & Access

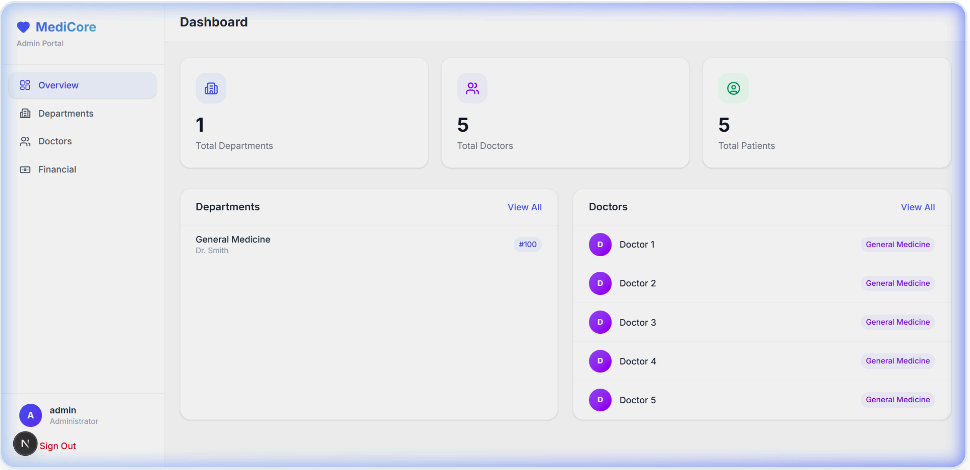
Technician Workspace

Restricted access for pathology staff to process diagnostic requests generated by doctors.

2 Key Features Displayed

- **Pending vs Completed Tabs:** Easy segregation of workload.
- **Request Details:** Instantly see test type, patient, referring doctor, and date.
- **Data Entry Portal:** 'Enter Report' button allows immediate upload of physical findings to the Oracle database.

Admin & Financials



1 Role & Access

System Oversight

Master control for hospital management to monitor departments, clear finances, and manage users.

2 Key Features Displayed

- **Global Statistics:** Total departments, doctors, and active patients.
- **Transaction Ledger:** Full financial history with transparency on hospital (20%) vs doctor (80%) shares.
- **Withdrawals Management:** Review and clear doctor payout requests securely.

What We Built & What We Learned

1 Project Achievements

Digital Platform

Complete end-to-end HMS replacing all paper workflows.

Oracle Database

9 normalized tables, PK/FK, sequences, triggers.

Secure Auth

JWT + bcrypt + bind variables for maximum security.

Financial Ledger

Automated 80/20 commission split management.

2 Challenges Faced & Overcame

3 Oracle 11g Configuration

Complex local setup and JDBC/OCI driver configuration required extensive troubleshooting.

4 PL/SQL Complexity

Writing stored procedures and triggers required deep understanding of Oracle's procedural SQL.

5 Complex Relationships

Designing 9 normalized tables with correct FK chains was the most critical design challenge.

Future Roadmap & Conclusion

1 Planned Future Improvements

- AI Diagnosis Support
- SMS Notifications
- Video Consultation
- Mobile Application

2 Conclusion

→ Complete HMS Delivered

Successfully digitized the full hospital workflow — appointments, prescriptions, labs, and financials.

→ Secure & Scalable

Oracle relational database with JWT-secured API — production-ready design patterns throughout.

Healthcare, Digitally Empowered.

Thank You — Questions?